

ASSESSMENT GUIDE

Unit Code: ITEC613 Machine Learning with Scikit-Learn 2025

Study Period, S1 2025

Assessment number (2)

Assessment Artefact: Python Codes and Comments

Weighting [30%]

Why this assessment?

- Opportunity to apply theory into practice
- Exposure to real-life scenario
- Enhance students' hand-on experience with python packages numpy, matplotlib, scikit-learn
- Establish the students' understanding of data preparation and manipulation using Python
- The feedback from this assessment will help students to be ready to correct any conceptual misunderstanding and apply in the next assessment

What are the types of employability skills that I will acquire upon completion of this assessment?

Skill Type	
Developed critical and analytical thinking	☒
Developed ability to solve complex problems	☐
Developed ability to work effectively with others	☐
Developed confidence to learn independently	☒
Developed written communication skills	☒
Developed spoken communication skills	☒
Developed knowledge in the field study	☒
Developed work-related knowledge and skills	☒

Assessment Overview:

Purpose, as written in the EUO

Due date:	09/05/2025, 23:59 pm AEST, on Friday of Week 9
Weighting:	30%
Length and/or format:	data exploring tasks
Learning outcomes assessed	LO2
Graduate attributes assessed	GA5, GA8

How to submit:	task submission – via CANVAS
Return of assignment:	Via CANVAS
Assessment criteria:	Rubric: see end of document

Context

Suppose the students take a Data Engineer role in a company. One of their daily duties would be processing huge amount of data for different projects. So this assignment will guide the students how to handle these situations with an example.

Instructions

MNIST number dataset a set of 70,000 small images of digits handwritten by high school students and employees of the US Census Bureau. Each image is labeled with the digit it represents. This set has been studied so much that it is often called the “hello world” of Machine Learning: whenever people come up with a new classification algorithm they are curious to see how it will perform on MNIST, and anyone who learns Machine Learning tackles this dataset sooner or later.

Instructions to explore this dataset are:

1. Data Acquisition and Initial Analysis:
 - Retrieve the MNIST dataset.
 - Perform exploratory data analysis to understand the dataset's structure, including
 - i. how many images
 - ii. how many features and the range of feature values (e.g., histogram of the data value), relating it to real-world, such as real images.
 - iii. how many categories/labels (discrete or continuous type) and what they are?
 - iv. visualize at least three randomly selected samples within each category (feel the variance of the data)
 - v. visualize more data samples to see whether there are bad data samples need to be removed. What bad data samples do you think can be?
2. Data Preparation and Manipulation:
 - Apply dimensionality reduction techniques (PCA and t-SNE) to the MNIST dataset and visualize the results.
 - Split the dataset into training (60,000 samples) and testing (10,000 samples) sets.
3. Machine Learning Model Implementation:
 - Train a Random Forest classifier on the original dataset and record its performance.
 - Use PCA to reduce the dataset's dimensionality to 174. Train a new Random Forest classifier on the reduced dataset and see how long it takes. Was training much faster? Then, evaluate the classifier on the test set. How does it compare to the previous classifier?
4. Critical Evaluation and Conclusion:
 - Provide a comprehensive evaluation of the performance of the models.
 - Summarize findings and insights.
5. Research Question: Explore how various image preprocessing methods (e.g., normalization, binarization, noise reduction, and image augmentation) influence the performance of at least two different machine learning models (e.g., Convolutional Neural Networks and Random Forest classifiers) trained on the MNIST dataset. Analyze the models' accuracy, training time, and ability to generalize to test data. Discuss your findings' implications for designing machine learning pipelines in digit recognition tasks.
6. Reflect on the composition and diversity of the MNIST dataset, considering its impact on the training process and model performance. Explore how the inclusion of a more diverse set of handwriting samples (e.g., different handwriting styles, inclusion of characters from non-Latin alphabets, or samples from wider age

groups) might affect the accuracy and generalizability of machine learning models trained for digit recognition tasks.

Submission Structure & Guidelines

Structure of the Submission

Please prepare a **Jupyter Notebook (.ipynb)** as your main submission file for this assessment.

Your notebook should be **well-organized** and structured to alternate between:

- **Text cells:** Use these to explain your approach, reasoning, and interpretation of results.
- **Code cells:** Contain your implementation and results of each task.

Each section must clearly correspond to the tasks listed in the assessment brief.

You are encouraged to refer to the textbook:

“Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow”

as a guide for completing the coding components.

How Do I Submit?

1. Prepare Your Submission

- Ensure your Jupyter Notebook (.ipynb) is complete with all required tasks, commentary, and outputs.
- Export your notebook as a **PDF** (File → Print → Save as PDF).

2. Access CANVAS

- Log in to your Canvas course page.
- Navigate to the **Assessment 2** section and open the submission link.

3. Upload Your Files

- Upload your .ipynb file and your .pdf version.
- You may include any supporting files (Word, text, or additional documentation).

4. Submit Your Work

- Carefully review your uploads for accuracy and completeness.
- Click **“Submit”** to finalize your submission.

! Academic Integrity Reminder

All code submissions will be automatically cross-checked via **Turnitin** against other students' work and online sources.

Ensure that your work is your own and that you properly explain and comment your code to demonstrate **your understanding**.

Failure to uphold academic integrity may result in serious academic consequences.

Submission checklist

I have formatted my report as per the specifications	☐
I have checked my Turnitin report and taken appropriate actions to ensure that the submission satisfies academic integrity (if applicable)	☐
I have actioned feedback advice provided to me from labs feedback (if applicable)	☐
I have submitted my work before the due date/time	☐
I have submitted feed forward template along with my assignment submission	☐

Note

This is a task for any instance of follow-on assignment (assessment 2 and 3). This must be submitted as the first page of the follow-on assignment (assessment 2 and 3) to ensure you acted on the feedback provided to you in the previous assignment (this is not counted as part of the assessment word count).

How did you act on the feedback?

Feedback is an important component of learning. Please consider the feedback you received in your last assignment and provide a response on how you acted on, or intend to act upon, that feedback, and how it has informed the current assignment task. Submit this sheet along with your assignment.

Questions	Your learning from the previous assignment feedback
How have you acted on the feedback from previous assignment to improve your work in this assignment?	(e.g. based on my previous feedback, I made sure that I supported my discussion, position, ideas, concepts with peer reviewed journal references in this assignment)
What is your expectation around the type of feedback that enhances your learning?	(e.g. I want to know where I made a mistake and how I can correct them and not make the same mistake again i.e. I want specific feedback that will help me to improve my learning and performance in the next assignment)
Did you have any difficulty understanding or acting on previous feedback? Please be	(e.g. feedback provided in my previous assignment was very generic I did not know how to improve my work. So, I would
as specific as possible so that you can gain further feedback/clarify anything you do not understand in the feedback	like the teacher to explain more on xxxx aspects of the feedback or I would like an opportunity to have a dialogue to understand the feedback)

Some Helpful Websites and Resources

Anaconda environment <https://docs.anaconda.com/anaconda/>

Python official website: <https://www.python.org/>

<https://numpy.org/>

<https://scikit-learn.org/stable/>

<https://pandas.pydata.org/>

<https://matplotlib.org/>

Who can help me?

Studiosity

Academic skills Unit (ASU)

Places –NLiC Mahmoud Bekhit (Mahmoud.bekhit@acu.edu.au)

I'm having problems

Special Consideration: This [form](#) is used by students to apply for **Special Consideration** for assessable work in

studies at Australian Catholic University. Approval of such applications will only be granted to students who are legitimately disadvantaged in their assessment due to exceptional and unforeseen **circumstances** beyond their control.

Referencing

All referencing should be in [ACU Harvard style](#); However if you are coming from another faculty, you may choose to use your usual referencing style. If this is the case you must indicate at the top of your reference list what referencing style you are using (e.g. APA, MLA, Chicago, etc).

Please ensure your assignment makes use of in-text citations and a reference list. Missing citations or references is equivalent to plagiarism.

Criteria

The full criteria is compiled in a rubric, which can be found on the following page/s.

Rubric for [ITEC613 AT2, 30%]

Relevant LO/GAs	Criterion (related to a single GA from the related LO – one GA per criterion)	Does not meet expectations	Meets expectations	Exceeds expectations		
		NN	PA	CR	DI	HD
GA5 LO2 Weight=15 marks TL=3 Learning stage = I and D	Demonstrate correct understanding of the data preparation and manipulation concepts	Fail to adequately demonstrate correct understanding of the data preparation and manipulation concepts including basic preprocessing and visualization techniques. (0 – 7.35)	Adequately demonstrate correct understanding of the data preparation and manipulation concepts. Basic tasks are completed satisfactorily, but with minimal exploration of the dataset's characteristics. (7.5 – 9.6)	Credibly demonstrate correct understanding of the data preparation and manipulation concepts. Shows ability to explore and visualize the dataset effectively, identifying key characteristics and potential issues. (9.75 – 11.1)	Distinctively demonstrate correct understanding of the data preparation and manipulation concepts. Includes effective use of preprocessing techniques and insightful dataset analysis, showing potential impacts on model performance. (11.25 – 12.6)	Highly distinctively demonstrate correct understanding of the data preparation and manipulation concepts. Provides deep insights into the dataset, effectively uses advanced preprocessing techniques, and critically evaluates their impact on machine learning models, addressing the research question on preprocessing techniques' impact on model accuracy. (12.75 – 15)
GA8 LO2 Weight=15 marks TL=3 Learning stage = I and D	Demonstrate python programming skills by implementing data preparation and manipulation codes with packages scikit-learn, numpy, matplotlib.	Fail to adequately demonstrate python programming skills by implementing data preparation and manipulation codes with packages scikit-learn, numpy, matplotlib. Lacks clarity in code and explanation.	Adequately demonstrate python programming skills by implementing data Preparation and manipulation codes with packages scikit-learn, numpy, matplotlib. Basic tasks are completed satisfactorily, but with limited efficiency or depth.	Credibly demonstrate python programming skills by implementing data preparation and manipulation codes with packages scikit-learn, numpy, matplotlib. Shows proficiency in using the mentioned packages and provides clear, efficient code with explanations.	Distinctively demonstrate python programming skills by implementing data preparation and manipulation codes with packages scikit-learn, numpy, matplotlib. Shows innovative use of the packages to achieve optimal results, including tackling the critical thinking question on the limitations of the MNIST dataset.	Highly distinctively demonstrate python programming skills by implementing data preparation and manipulation codes with packages scikit-learn, numpy, matplotlib. Showcases exceptional ability to address complex challenges, including the research question and critical thinking aspect, through innovative solutions and critical analysis of outcomes.

		(0 – 7.35)	(7.5 – 9.6)	(9.75 – 11.1)	(11.25 – 12.6)	(12.75 – 15)
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